THE WRONSKIAN

# Testing $e^x$, $e^{2x}$, $e^{3x}$ for independence

A Wronskian that is nonzero at some point proves the functions linearly independent; this one is nonzero everywhere.

01

**WRONSKIAN MATRIX** — BEFORE

$$W = \det \begin{pmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{pmatrix}$$

↓ Factor each column — $e^x$ from column 1, $e^{2x}$ from column 2, $e^{3x}$ from column 3

02

**VANDERMONDE MATRIX, NODES 1, 2, 3**

$$W = e^{6x} \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

↓ Vandermonde determinant — $(2-1)(3-1)(3-2) = 2$

03

NEVER ZERO

AFTER

$$W = 2e^{6x} \neq 0$$

▼ A ZERO WOULD NOT PROVE DEPENDENCE